

2110. Number of Smooth Descent Periods of a Stock

Difficulty: Medium

Link: <https://leetcode.com/problems/number-of-smooth-descent-periods-of-a-stock/?envType=daily-question&envId=2025-12-15>

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Problem:

You are given an integer array `prices` representing the daily price history of a stock, where `prices[i]` is the stock price on the i^{th} day.

A **smooth descent period** of a stock consists of **one or more contiguous** days such that the price on each day is **lower** than the price on the **preceding day** by **exactly** `1`. The first day of the period is exempted from this rule.

Return *the number of smooth descent periods*.

Example 1:

Input: `prices = [3,2,1,4]`

Output: 7

Explanation: There are 7 smooth descent periods:

`[3]`, `[2]`, `[1]`, `[4]`, `[3,2]`, `[2,1]`, and `[3,2,1]`

Note that a period with one day is a smooth descent period by the definition.

Example 2:

Input: prices = [8,6,7,7]

Output: 4

Explanation: There are 4 smooth descent periods: [8], [6], [7], and [7]

Note that [8,6] is not a smooth descent period as $8 - 6 \neq 1$.

Example 3:

Input: prices = [1]

Output: 1

Explanation: There is 1 smooth descent period: [1]

Constraints:

- $1 \leq \text{prices.length} \leq 10^5$
- $1 \leq \text{prices}[i] \leq 10^5$